Filter Design for Signal Frequency Components

Task:

Suggest appropriate filters to isolate the following frequency components from the input signal. You will need to decide on the filter type (Low Pass, High Pass, Band Pass, or Band Stop) and specify the cutoff frequency or frequencies.

Input Signal: The input signal is a sum of sine waves with the following frequencies

$$Input\ Signal = A_1 \cdot sin\ (2\pi \cdot 100 \cdot t) + A_2 \cdot sin\ (2\pi \cdot 200 \cdot t) + A_3 \cdot sin\ (2\pi \cdot 300 \cdot t) + A_4 \cdot sin\ (2\pi \cdot 400 \cdot t)$$

Frequency Component	**Filter Type**	**Cutoff Frequency/Frequencies**
100 Hz	LPF	110 Hz
400 Hz	HPF	390 Hz
100 Hz and 200 Hz	LPF	210 Hz
200 Hz	BPF	Passband: 190-210 Hz
300 Hz	BPF	Passband: 290-310 Hz
300 Hz and 400 Hz	HPF	290 Hz
200 Hz and 300 Hz	BPF	190-310 Hz
200 Hz, 300 Hz, and 400 Hz	HPF	190 Hz
100 Hz and 400 Hz	BSF	Stopband: 110-390 Hz